



③ PDA - stack, tape finite length
read/write, finite control unit, →

Computer Science and Engineering

Continuous Assessment Test – II

push/pop operations

Regulations – R2021

TM - infinite length tape, read head, FC unit; head $\rightarrow$, read & write & memorize operations.

Degree & Branch	B.E. & CSE				Semester	VI
Subject Code & Name	UCS2603 & Theory of Computation					
Academic Year	2024-2025 EVEN	Batch	2022-2026	Date	26.03.2025	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2	Identify relationship between finite automata and regular expressions (K3)
CO3	Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)
CO4	Construct Turing machine for a given language (K3)
CO5	Explain the decidability or undecidability of various problems and make use of it (K3)

$P \mid C L \geq n \quad z = uvwxy$ pumping lemma $uv^iwx^iy^j \in L, i \geq 0$
 $u, v, w, x, y \neq \epsilon, |v| \geq 1, |y| \geq 1$
 $\leq n$

Part – A ($4 \times 2 = 8$ Marks)

		KL	CO	PI
1.	Show the resultant grammar after removing the <i>useless symbols</i> . $S \rightarrow AB \mid CA, B \rightarrow BC \mid AB, A \rightarrow a, C \rightarrow aB \mid b$	K2	CO3	1.4.1
2.	Define Pumping Lemma of Context Free Language (CFL) with its application. $\rightarrow$ given language is CFL or not.	K1	CO3	1.4.1
3.	Compare Push Down Automata (PDA) with Turing Machine (TM) with respect to its design and operation.	K2	CO4	1.4.1
4.	How can Recursive (R) and Recursively Enumerable (RE) languages be defined? $\downarrow$ Turing decidable $\downarrow$ Turing acceptable or semi decidable.	K1	CO5	1.3.1

Part – B ($3 \times 6 = 18$ Marks)

		KL	CO	PI
5.	Consider the grammar: $S \rightarrow iCtS, S \rightarrow iCtSeS, S \rightarrow a$ and $C \rightarrow b$, where “i”, “t” and “e” stands for “if”, “then”, and “else”, and “C” and “S” for “conditional” and “statement” respectively. Construct a leftmost and rightmost derivations for the sentence $w = \text{“ibtibtaea”}$ with parse trees.	K3	CO3	2.1.3

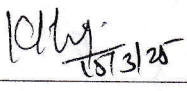
① $\rightarrow$ Generating $\{S, A, C\} \rightarrow$ variable B & its productions are removed.

② Reachable from start symbol $S \rightarrow \{S, A, C\}$
 Ans: $S \rightarrow CA$
 $A \rightarrow a$
 $C \rightarrow b$

6.	Apply the procedure of CFG to PDA, to find an equivalent PDA that accepts on empty stack. $S \rightarrow aB \mid bA$ $A \rightarrow a \mid aS \mid bAA$ $B \rightarrow b \mid bS \mid aBB$	K3	CO3	2.4.1
CFG to PDA Rules (2) conversion (4)				
7.	Define Greibach Normal Form (GNF) and construct GNF for the following Context Free Grammar G. $S \rightarrow AB$ $A \rightarrow BS \mid b$ $B \rightarrow SA \mid a$	K3	CO4	1.4.1
2+4 (lemma + conversion)				

Part – C (2 × 12 = 24 Marks)

		KL	CO	PI
8.	Construct a PDA for the following languages: (5 + 7 marks) (a) Accepts by final state for even length palindromes where $\Sigma = \{0, 1\}$ and validate the string "1001" using ID. (b) Accepts by empty stack for $L = \{a^n b^{3n} \mid n \geq 1\}$ where $\Sigma = \{a, b\}$ and validate the string "aabbabbbb" using ID.	K3	CO3	2.1.3, 13.3.1, 13.4.1
(OR)				
9.	Construct a CFG for the given PDA $M = (\{q_0, q_1\}, \{0, 1\}, \{X, Z\}, \delta, q_0, Z, \phi)$ where δ is given by: $\delta(q_0, 0, Z) = (q_0, XZ)$ $\delta(q_0, 0, X) = (q_0, XX)$ $\delta(q_0, 1, X) = (q_1, Z)$ $\delta(q_1, 1, X) = (q_1, \lambda)$ $\delta(q_1, \lambda, X) = (q_1, \lambda)$ $\delta(q_1, \lambda, Z) = (q_1, \lambda)$	K3	CO3	2.1.3, 13.3.1, 13.4.1
PDA to CFG rules (2) conversion (10)				
10.	Construct a TM to perform the subtraction function and compute the subtraction of $w = "0010"$ $f(m, n) = m - n$ if $m > n$ $f(m, n) = B$ if $m \leq n$	K3	CO4	2.1.3, 13.3.1, 13.4.1
(OR)				
11.	Construct a TM for the given language $L = \{a^n b^n c^n \mid n \geq 1\}$ and validate the string $w = "aabbcc"$.	K3	CO4	2.1.3, 13.3.1, 13.4.1
(10+2)				

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Course Coordinator	PAC Team	HOD

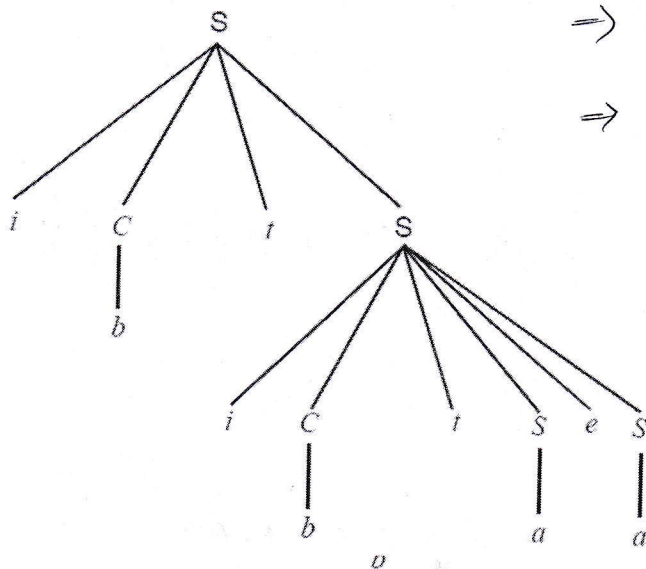
(3)

5. LMD and RMD note: The given grammar is ambiguous also
 $w = ibtibtaea$ Since it has 2 distinct LMD or RMD derivations

Leftmost derivation 1:

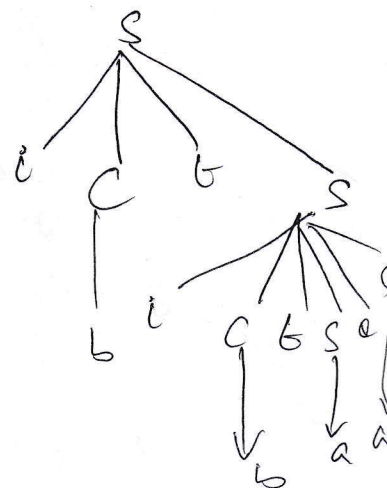
$S \Rightarrow iCtS$
 $\Rightarrow ibtS \ (C \rightarrow b)$
 $\Rightarrow ibtiCtSeS \ (S \rightarrow iCtSeS)$
 $\Rightarrow ibtibtSeS \ (C \rightarrow b)$
 $\Rightarrow ibtibtaeS \ (S \rightarrow a)$
 $\Rightarrow ibtibtaea \ (S \rightarrow a)$

(2)



RMD 1:

$S \Rightarrow iCtS$
 $\Rightarrow iCtiCtSeS \ (S \rightarrow iCtSeS)$
 $\Rightarrow iCtiCtSea \ (S \rightarrow a)$
 $\Rightarrow iCtibtaea \ (S \rightarrow a)$
 $\Rightarrow ibtibtaea \ (C \rightarrow b)$

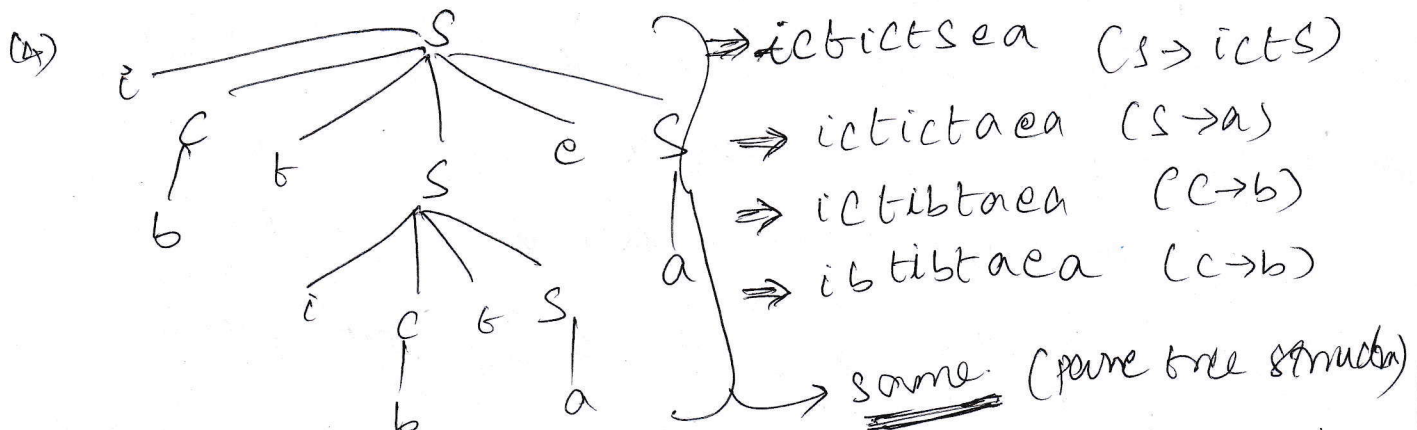


(3) Leftmost derivation 2:

$S \Rightarrow iCtSeS$
 $\Rightarrow ibtSeS \ (C \rightarrow b)$
 $\Rightarrow ibtiCtSeS \ (S \rightarrow iCtS)$
 $\Rightarrow ibtibtSeS \ (C \rightarrow b)$
 $\Rightarrow ibtibtaeS \ (S \rightarrow a)$
 $\Rightarrow ibtibtaea \ (S \rightarrow a)$

RMD 2:

$S \Rightarrow iCtSeS$
 $\Rightarrow iCtSeiCtS \ (S \rightarrow iCtS)$
 $\Rightarrow iCtSeiCta \ (S \rightarrow a)$
 $\Rightarrow iCtSea \ (S \rightarrow a)$



6. CFG to PDA

The PDA $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ where

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{S, A, B, z_0\}$$

$$F = \{q_2\}$$

q_0 - startstate.

1. First the start symbol S is put on the stack.

$$\delta(q_0, \lambda, z_0) = \{(q_1, Sz_0)\}$$

2. The production $S \rightarrow aB \mid bA$ is simulated by PDA as per $A \rightarrow \alpha$ rule. $\delta(q_1, \lambda A) = \{(q_1, \alpha)\}$

$$\delta(q_1, \lambda, S) = \{(q_1, aB), (q_1, bA)\}$$

3. In a similar manner, the other productions are :

$$\delta(q_1, \lambda, A) = \{(q_1, a), (q_1, aS), (q_1, bAA)\}$$

$$\delta(q_1, \lambda, B) = \{(q_1, b), (q_1, bS), (q_1, aBB)\}$$

4. For each $a \in \Sigma$ the corresponding productions are : $\delta(q_1, a, a) = \{(q_1, \lambda)\}$

$$\delta(q_1, a, a) = \{(q_1, \lambda)\}$$

$$\delta(q_1, b, b) = \{(q_1, \lambda)\}$$

5. Then the end of the derivation is identified by the stack symbol

$$\delta(q_1, \lambda, z_0) = \{(q_2, \lambda)\}$$

(7) GNF: $\left[A \rightarrow a \text{ (or)} A \rightarrow \alpha \alpha \mid \alpha \in V^* \right] \left[S = A_1 \mid A = A_2 \mid B = A_3, \text{ check for lemma 1 \& lemma 2 of GNF} \right]$

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1 \mid b$$

$$A_3 \rightarrow A_1 A_2 \mid a \rightarrow \text{lemma 2 applicable.}$$

Lemma 2: $A \rightarrow A\alpha_1 \mid A\alpha_2 \mid \dots \mid A\alpha_r \mid \beta_1 \mid \beta_2 \mid \dots \mid \beta_m$ then

$$A \rightarrow \beta_1 \mid \beta_2 \mid \dots \mid \beta_m \text{ \& } A \rightarrow \beta_1 B \mid \beta_2 B \mid \dots \mid \beta_m B$$

$$B \rightarrow \alpha_1 \mid \alpha_2 \mid \dots \mid \alpha_r \text{ \& } B \rightarrow \alpha_1 B \mid \alpha_2 B \mid \dots \mid \alpha_r B$$

Soln: ① Substitute A_1 in A_3 \& A_1, A_2 remains same.

$$A_3 \rightarrow A_2 A_3 A_2 \mid a \text{ substitute } A_2 \text{ in } A_3.$$

$$\frac{A_3 \rightarrow A_3 A_1 A_3 A_2}{A} \mid \frac{b A_3 A_2}{\beta_1} \mid \frac{a}{\beta_2} \text{ apply lemma 2.}$$

$$A_3 \rightarrow b A_3 A_2 \mid a \mid b A_3 A_2 B_3 \mid a B_3 \quad B_3 \rightarrow A_1 A_3 A_2 \mid A_1 A_3 A_2$$

5

② substitute A_3 in A_2

$$A_2 \rightarrow bA_3A_2A_1 \mid aA_1 \mid bA_3A_2B_3A_1 \mid aB_3A_1 \mid b$$

③ substitute A_2 in A_1

$$A_1 \rightarrow bA_3A_2A_1A_3 \mid aA_1A_3 \mid bA_3A_2B_3A_1A_3 \mid aB_3A_1A_3 \mid bA_3$$

④ substitute A_1 in B_3

$$B_3 \rightarrow bA_3A_2A_1A_3A_3A_2 \mid aA_1A_3A_3A_2 \mid bA_3A_2B_3A_1A_3A_3A_2 \mid aB_3A_1A_3A_3A_2 \mid bA_3A_3A_2A_2$$

$$B_3 \rightarrow bA_3A_2A_1A_3A_3A_2B_3 \mid aA_1A_3A_3A_2B_3 \mid bA_3A_2B_3A_1A_3A_3A_2B_3 \mid aB_3A_1A_3A_3A_2B_3 \mid bA_3A_3A_2B_2$$

Total productions in GRNF: $A_3 - (4)$, $A_2 - (5)$, $A_1 - (5)$
 $B_3 - (10) = (24)$ productions.

8. (a) $L = \{ww^R \mid w \in \{0,1\}^+\}$

$$\delta(q_0, 0, 2_0) = (q_0, 0, 2_0)$$

$$\delta(q_0, 1, 2_0) = (q_0, 1, 2_0)$$

$$\delta(q_0, 0, 1_0) = (q_0, 0, 1_0)$$

$$\delta(q_0, 0, 1_1) = (q_0, 0, 1_1)$$

$$\delta(q_0, 1, 1_0) = (q_0, 1, 1_0)$$

$$\delta(q_0, 1, 1_1) = (q_0, 1, 1_1)$$

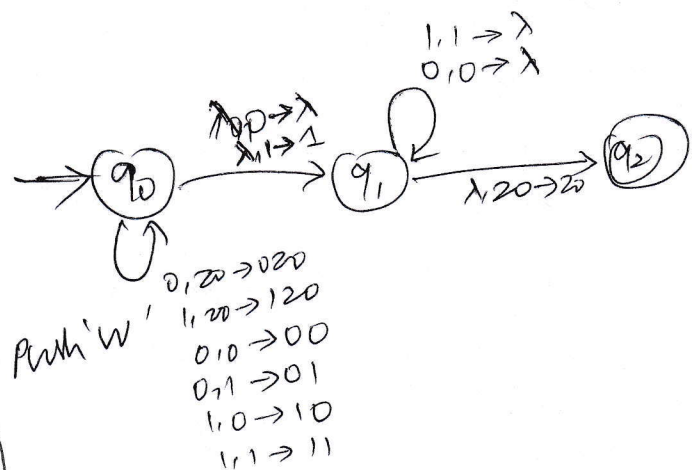
$$\delta(q_0, \lambda, 1_0) = (q_1, 0)$$

$$\delta(q_0, \lambda, 1_1) = (q_1, 1)$$

$$\delta(q_1, 0, 1_0) = (q_1, 0, 1_0)$$

$$\delta(q_1, 1, 1_1) = (q_2, 2_0)$$

$$\delta(q_1, \lambda, 2_0) = (q_2, 2_0)$$



an arrow is between w and w^R

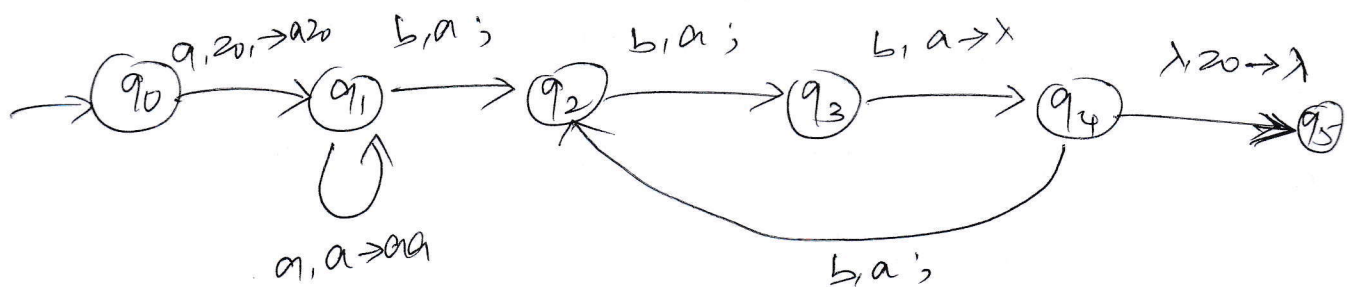
Finally define m .

$$\delta(q_0, w, 1_0, 2_0)$$

$$\vdash (q_2, 2_0)$$

$w = w^R$
 $q_2 \in F$

5. (b) $L = \{a^n b^{3n} \mid n \geq 1\}$



$$\delta(q_0, aabbbbbb, z_0) \vdash^* (q_5, \lambda) \text{ (or) } (q_4, \lambda)$$

9. PDA to CFG $G = (V, T, P, S)$

$$V = [QAP] \quad \forall q, p \in Q \text{ \& } A \in \Gamma \cup \{z_0\}$$

$$T = \Sigma \quad S = [q_0 z_0 q]$$
 for each $q \in Q$.

δ : Push Rule $\delta(q, a, A) = (q_{mH}, \beta_1, \beta_2, \dots, \beta_m)$

$$[q A q_{mH}] \rightarrow a [q, \beta_1, q_2] [q_2 \beta_2 q_3] \dots [q_m \beta_m q_{mH}]$$

Pop Rule $\delta(q, a, A) = (q_1, \lambda)$

$$[q A q_1] \rightarrow a$$

① $\delta(q_0, 0, z) = (q_0, xz)$

$$[q_0, z, q_0] \rightarrow 0 [q_0, x, q_0] [q_0, z, q_0]$$

$$[q_0, z, q_0] \rightarrow 0 [q_0, x, q_1] [q_1, z, q_0]$$

$$[q_0, z, q_1] \rightarrow 0 [q_0, x, q_0] [q_0, z, q_1]$$

$$[q_0, z, q_1] \rightarrow 0 [q_0, x, q_1] [q_1, z, q_1]$$

② $\delta(q_0, 0, x) = (q_0, xx)$

$$[q_0 x q_0] \rightarrow 0 [q_0 x q_0] [q_0 x q_0]$$

$$[q_0 x q_0] \rightarrow 0 [q_0 x q_1] [q_1 x q_0]$$

$$[q_0 x q_1] \rightarrow 0 [q_0 x q_0] [q_0 x q_1]$$

$$[q_0 x q_1] \rightarrow 0 [q_0 x q_1] [q_1 x q_1]$$

$$(3) \delta(q_0, 1, x) = (q_1, z)$$

$$[q_0 \times q_0] \rightarrow 1 [q_1 \neq q_0]$$

$$[q_0 \wedge q_1] \rightarrow 1 [q_1 \neq q_1]$$

$$(4) \delta(q_1, 1, x) = (q_1, \lambda)$$

$$[q_1 \times q_1] \rightarrow 1$$

$$(5) \delta(q_1, \lambda, x) = (q_1, \lambda)$$

$$[q_1 \times q_1] \rightarrow \lambda$$

$$(6) \delta(q_1, \lambda, z) = (q_1, \lambda)$$

$$[q_1 \neq q_1] \rightarrow \lambda$$

$$(10) \boxed{0|0|0|2|0|0} \text{ iff } m > n$$

$$\boxed{B|B|0|B|B|B} \text{ o/p}$$

$$\boxed{0|0|1|0|0}$$

$$\boxed{0|1|0|0}$$

$$\boxed{B|B|B|B|B} \text{ o/p}$$

$$\text{iff } m \leq n$$

Forward (matching m...0's with n...0's)

$$\delta(q_0, 0) = (q_1, B, R)$$

$$\delta(q_1, 0) = (q_1, 0, R)$$

$$\delta(q_1, 1) = (q_2, 1, R)$$

$$\delta(q_2, 1) = (q_2, 1, R)$$

$$\delta(q_2, 0) = (q_3, 1, L)$$

Backward (to repeat loop)

$$\delta(q_3, 1) = (q_3, 1, L)$$

$$\delta(q_3, 0) = (q_3, 0, L)$$

$$\delta(q_3, B) = (q_0, B, R)$$

Termination $m > n$

$$\delta(q_2, B) = (q_4, B, L)$$

$$\delta(q_4, 1) = (q_4, B, L)$$

$$\delta(q_4, 0) = (q_4, 0, L)$$

$$\delta(q_4, B) = (q_6, 0, H)$$

(one extra symbol changed is modified to original input)

$$\delta(q_0, 0010B) \xrightarrow[\text{steps.}]{+} B B q_6 0 B B$$

Termination $m \leq n$

$$\delta(q_0, 1) = (q_5, B, R)$$

$$\delta(q_5, 1) = (q_5, B, R)$$

$$\delta(q_5, 0) = (q_5, 0, R)$$

$$\delta(q_5, B) = (q_6, B, H)$$

$$\boxed{q_6 \in F}$$

(8)

① $L = \{a^n b^n c^n \mid n \geq 1\}$ $w = aabbbc$

Forward step: matching a with b with c

$$\left. \begin{array}{l} \textcircled{1} \delta(q_0, a) = (q_1, x, R) \\ \textcircled{2} \delta(q_1, a) = (q_1, a, R) \\ \textcircled{3} \delta(q_1, b) = (q_2, y, R) \\ \textcircled{4} \delta(q_2, b) = (q_2, b, R) \\ \textcircled{5} \delta(q_2, c) = (q_3, z, R) \end{array} \right\}$$

Crossing processed
a's & b's.

$$\left. \begin{array}{l} \textcircled{6} \delta(q_1, y) = (q_1, y, R) \\ \textcircled{7} \delta(q_2, z) = (q_2, z, R) \end{array} \right\}$$

Reverse step:

$$\begin{array}{l} \textcircled{8} \delta(q_3, b) = (q_3, b, L) \\ \textcircled{9} \delta(q_3, z) = (q_3, z, L) \\ \textcircled{10} \delta(q_3, y) = (q_3, y, L) \\ \textcircled{11} \delta(q_3, a) = (q_3, a, L) \\ \textcircled{12} \delta(q_3, x) = (q_0, x, R) \rightarrow \text{loop from } q_0 \end{array}$$

Termination condition:

$$\begin{array}{l} \textcircled{13} \delta(q_0, y) = (q_4, y, R) \\ \textcircled{14} \delta(q_4, y) = (q_4, y, R) \\ \textcircled{15} \delta(q_4, z) = (q_5, z, R) \\ \textcircled{16} \delta(q_5, z) = (q_5, z, R) \\ \textcircled{17} \delta(q_5, B) = (q_6, B, H) \end{array}$$

9

$\delta(q_0, aabbbc)$ (25) steps.

$\vdash xq_1abbbc$

$\vdash xaq_2bbcc$

$\vdash xayq_2bcc$

$\vdash xay\bar{b}q_2cc$

$\vdash xayq_3bzc$

$\vdash xaq_3ybzc$

$\vdash xq_3aybzc$

$\vdash q_3xaybzc$

$\vdash xq_0aybzc$ (looping)

$\vdots$

$\vdash xxq_0yyzzB$

$\vdash xxq_4yzzB$

$\vdots$

$\vdash xxyyq_4zzB$

$\vdash xxyyzq_5zB$

$\vdash xxyyzzq_5B$

$\vdash xxyyzzq_6B$ (Final state)